

From Backtest Output to Auditable Evidence

A Confirmatory Point-in-Time Replication of A-Share Factors

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ABSTRACT

This working paper asks how much apparently favorable A-share factor evidence changes when a single research design is moved from a naive backtest toward point-in-time and implementation-aware measurement. A repository-locked specification evaluates return on equity, 60-session momentum, 20-session low volatility, and a fixed-weight composite across four cumulative variants: a final-survivor baseline (M0), a point-in-time listing universe (M1), publication-date alignment for accounting data (M2), and an audited implementation variant with tradability filters and a one-session lag (M3). The out-of-sample window contains 18 monthly observations from January 2025 through June 2026 and averages more than 4,300 securities per cross-section. All 16 planned factor-by-variant cells are reported. Publication-time alignment reduces mean ROE rank information coefficient from 0.0461 in M1 to 0.0105 in M2 and reduces composite IC from 0.0332 to 0.0038. Under M3, both become negative. Momentum remains negative in IC terms while its top-quintile-minus-universe spread is positive; low volatility exhibits the reverse sign pattern, underscoring that a single summary statistic is not an investment claim. The contribution is therefore evidentiary rather than promotional: a hash-bound receipt connects the data declaration, locked plan, code commit, and complete result matrix, while explicit claim gates rule out trading-readiness or generalization claims.

Keywords: A-shares; point-in-time data; publication lag; survivorship bias; factor replication; research governance

Research-use disclaimer. The study reports descriptive factor-return evidence, not an investable portfolio, recommendation, or claim of expected performance.

1. Introduction

Open-source quantitative projects often look complete before their empirical claims are complete. A repository may contain data adapters, strategy factories, backtest engines, walk-forward utilities, reports, and tests, yet still lack one object that a reader can independently audit: a fully specified real-market out-of-sample result whose data identity, timing assumptions, code state, and complete outcome matrix are bound together. This distinction matters because software quality and empirical credibility solve different problems. Passing tests can show that code behaves as specified; it cannot show that the chosen specification was not selected after viewing many alternatives or that the inputs were available at the historical decision time.

This paper converts one public A-share quantitative research repository from a broad engineering demonstration into a narrow empirical working paper. The question is deliberately mechanical: how much do survivorship controls, publication-time alignment, and an implementation lag change out-of-sample factor statistics? The exercise does not attempt to discover a new anomaly. It uses familiar signals—return on equity, price momentum, low volatility, and a pre-weighted composite—because their familiarity makes the data-timing audit easier to interpret. If a familiar signal weakens once the information set is reconstructed more conservatively, the result is useful even when no profitable strategy remains.

The design compares four cumulative variants. M0 admits only securities that survive in the final stock master and treats an accounting observation as available from its report-period end. M1 reconstructs listing membership by observation date. M2 retains that point-in-time universe but delays accounting availability until the vendor-provided publication date. M3 adds same-day tradability screens, a 20-session average-amount floor, and a one-session implementation lag. The comparison is not intended as a tournament. Every factor is carried through every variant, producing a 4×4 matrix specified before the confirmatory run.

The main empirical pattern is sharp. Point-in-time universe membership alone changes the results modestly, but publication-time alignment materially attenuates the accounting-based signals. Mean ROE rank IC falls from 0.0461 in M1 to 0.0105 in M2, a 77.2% decline relative to M1. Composite IC falls from 0.0332 to 0.0038, an 88.5% decline. In M3, both means turn negative. These estimates are based on only 18 monthly cross-sections, so the reported Newey–West statistics are descriptive diagnostics rather than declarations of statistical significance.

Two additional findings discipline interpretation. First, the price-only factors are unchanged from M1 to M2 by construction, because publication dates affect only fundamentals. That invariance acts as a useful internal control on the implementation. Second, the sign of average rank IC does not always match the

sign of the top-quintile-minus-universe mean return. Momentum has negative IC but positive top-quintile spread, while low volatility has positive IC but negative spread. This discordance can arise because a full-rank association and an extreme-bin contrast weight the cross-section differently, and because the paper averages statistics across dates rather than pooling observations. It also makes clear why one favorable number should not be promoted as portfolio evidence.

The paper contributes in three ways. Empirically, it quantifies the effect of successive information-set controls in a large A-share cross-section. Methodologically, it treats the variant sequence as an audit trail, so that each transition has a defined interpretation. Operationally, it records the study in a canonical receipt containing raw-file hashes, the plan hash, the bound analysis commit, sample counts, all planned results, and claim gates. The receipt is not a substitute for redistributable raw data, but it prevents local reports from silently disagreeing about the authoritative research state.

The contribution is intentionally bounded. The public repository does not contain the licensed real-market export, and the available public QData interface does not establish a complete historical vintage or as-of revision archive. Therefore this paper studies listing membership, publication-time availability, and execution timing; it does not study the history of subsequent data revisions. Nor does it estimate turnover, transaction costs, capacity, portfolio constraints, or live performance. The correct conclusion is that the evidence chain is stronger and the naive accounting results are fragile—not that the repository is portfolio-ready.

The remainder of the paper reviews related factor and replication research, describes the proprietary research export and its evidence boundary, states the locked design, reports the complete matrix, explains the governance mechanism, and closes with limitations and a prospective research agenda.

2. Related Literature and Research Expectations

2.1 Factor signals and the A-share setting

The cross-sectional asset-pricing literature has long linked expected returns to firm characteristics and past prices. Fama and French (1992) organize early U.S. evidence around size and book-to-market. Jegadeesh and Titman (1993) document intermediate-horizon return continuation, providing the canonical reference for momentum. Ang, Hodrick, Xing, and Zhang (2006) show that stocks with high idiosyncratic volatility earn anomalously low average returns, motivating defensive or low-volatility signals. Novy-Marx (2013) demonstrates that profitability contains information distinct from value. The present study does not reproduce those exact definitions; it uses ROE, 60-session price momentum, and the negative of 20-session realized volatility as transparent repository-level proxies.

China's equity market requires local attention rather than mechanical transfer of U.S. conventions. Liu, Stambaugh, and Yuan (2019) emphasize features of the Chinese market and develop size and value factors tailored to that setting. The current paper likewise focuses on the A-share investable universe, but its objective differs: it is not a new factor-model proposal. It asks whether a given signal measurement survives progressively stricter reconstruction of what was knowable and tradable at each historical date.

2.2 Replication, data snooping, and implementation

A large menu of strategies creates a selection problem even when each backtest is mechanically correct. White (2000) formalizes the danger of data snooping when the best rule is chosen from many alternatives. Harvey, Liu, and Zhu (2016) argue that the proliferation of tested factors requires more demanding evidentiary thresholds. Hou, Xue, and Zhang (2020) show that many published anomalies weaken or fail under standardized replication choices. McLean and Pontiff (2016) document lower post-publication returns for published predictors, consistent with both statistical bias and market learning. These studies motivate the paper's no-best rule: the confirmatory object is the full pre-specified matrix, not its most attractive cell.

Implementation assumptions can be as consequential as signal definitions. Novy-Marx and Velikov (2016) show that trading costs reshape anomaly economics. Brown, Goetzmann, Ibbotson, and Ross (1992) establish that excluding nonsurvivors biases performance evidence. This paper tackles only a subset of implementation concerns: historical listing membership, publication timing, coarse tradability screens, and a one-session lag. The result remains a factor-return study rather than a costed portfolio simulation.

The research expectations are directional rather than formal preregistered hypotheses. First, replacing final survivors with point-in-time membership should reduce bias when failed or delisted firms differ systematically from survivors. Second, aligning accounting information to publication dates should weaken ROE-based evidence if report-period dates reveal future information. Third, adding tradability filters and an implementation lag should generally make results more conservative, although the combined M2-to-M3 transition cannot identify which component causes any change. Fourth, results should be judged jointly across IC and extreme-bin spreads, not by whichever metric looks best ex post.

3. Data and Evidence Boundary

3.1 Research export

The confirmatory run uses a local Investoday historical A-share research export. The vendor permits local research use but not redistribution, so the raw files are absent from the public repository. According to the canonical receipt, the daily market panel spans 3 January 2023 through 24 July 2026, contains 3,894,242 rows, and covers 4,735 symbols. The study window uses 2023–2024 for training-period availability and 1 January 2025 through 30 June 2026 for out-of-sample evaluation. Eighteen monthly rebalance observations meet the locked minimum of 16.

Daily quote fields include date, symbol, vendor-adjusted close, turnover amount, ST status, and suspension status. The stock master supplies list and delist dates as well as final membership. Fundamental records include symbol, ROE, report-period end, and publication date. The design treats these fields as vendor observations. It does not independently reconstruct corporate-action adjustments, exchange notices, or original filings.

The receipt records SHA-256 hashes for each raw input, a hash of the study plan, the bound Git commit, sample counts, execution metadata, and the complete result matrix. Those hashes establish identity: a reader with authorized access to the same files can determine whether the inputs match the research run. They do not prove that the vendor data are correct, nor do they make the proprietary records independently available.

3.2 What point-in-time means here

Point-in-time is used narrowly. For universe membership, a security is eligible only after its listing date and before its delisting date. For accounting information, M2 and M3 use `publicationDate` to prevent a report from entering the cross-section before the vendor records it as published. For implementation, M3 observes the signal and screens on the rebalance session, then measures the forward return from the next session.

The study does not possess a complete vintage database showing every later correction to every historical field. Consequently, `publicationDate` alignment is not evidence about data-revision history. A value may have been corrected after first publication without that correction being represented as a distinct vintage in the export. The separate open QData interface offers versioned snapshot concepts, but its public boundary does not currently justify a claim that the present experiment reconstructs a complete as-of revision archive. Throughout this paper, ‘point-in-time’ therefore refers to membership and first-availability controls, not complete vintage replay.

3.3 Sample construction

Table 1. Locked sample and measurement choices

Item	Locked choice	Interpretation
Market panel	3 Jan 2023–24 Jul 2026	3,894,242 daily rows; 4,735 symbols
Training interval	1 Jan 2023–31 Dec 2024	Used to establish rolling histories and pre-OOS availability
Out-of-sample interval	1 Jan 2025–30 Jun 2026	18 first-session-of-month observations
Forward horizon	20 trading sessions	M3 begins one session after signal formation
Liquidity floor	CNY 5 million	20-session mean amount; M3 only
Minimum universe	3,000 symbols	All reported cells exceed the threshold

Note. Dates, thresholds, and counts are taken from the locked plan and canonical receipt. The first observed market session of each month is the rebalance date.

4. Confirmatory Design

4.1 Signals and outcomes

At each monthly rebalance, the program forms cross-sectional signals from data available under the relevant variant. ROE is the most recently available reported return on equity. Momentum is the 60-session percentage change in adjusted close. Low volatility is the negative of the rolling 20-session standard deviation of daily returns, so a larger signal value denotes lower volatility. The composite is fixed before the run as 50% of the ROE percentile rank, 30% of the momentum percentile rank, and 20% of the low-volatility percentile rank. No weights are optimized in the out-of-sample period.

For M0–M2, the 20-session forward return is the adjusted close 20 sessions after the rebalance close divided by the rebalance close, minus one. For M3, the denominator is shifted to the next session and the endpoint is shifted correspondingly, representing a one-session implementation lag. These are raw security returns. No benchmark hedge, sector neutralization, weighting rule, rebalance turnover, fee, tax, price limit, or capacity model is applied.

The primary association measure is the monthly cross-sectional correlation between the rank of the signal and the rank of the forward return. The paper calls its time-series average mean rank IC. The second measure sorts securities by signal and subtracts the equal-weight universe mean return from the equal-weight return of the top signal quintile. The latter is a diagnostic contrast, not a self-financing long-short portfolio. Each row also reports the mean cross-sectional sample size.

For the time series of 18 monthly IC observations, the code reports a Newey–West t-statistic with lag three, following the heteroskedasticity and autocorrelation consistent covariance approach of Newey and West (1987). With such a short time series, asymptotic calibration is weak. The t-statistics are retained because they were part of the locked output contract, but the discussion treats them as measures of stability rather than binary significance tests.

4.2 Cumulative audit variants

Table 2. Variant definitions and identifying interpretation

Variant	Label	Cumulative rules	Clean comparison
M0	Naive baseline	Final-survivor universe; fundamentals available at report-period end; same-close horizon	Shows the optimistic repository baseline
M1	PIT universe	Historical list/delist membership; report-period timing retained; same-close horizon	Isolates membership reconstruction relative to M0
M2	PIT publication	M1 plus accounting availability at publicationDate	Isolates publication timing for ROE and composite
M3	Audited lag	M2 plus ST/suspension exclusions, CNY 5m amount floor, and one-session lag	Combined implementation stress; not separately identified

Note. M1→M2 is a clean timing comparison for accounting-dependent signals. M2→M3 changes several implementation assumptions simultaneously and must not be decomposed causally.

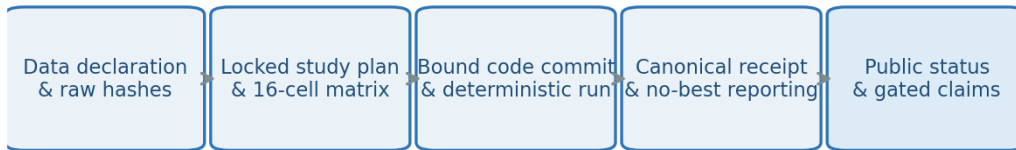
4.3 Locking, completeness, and claim gates

The plan fixes the question, factor definitions, composite weights, date intervals, four variants, output metrics, minimum sample requirements, and the instruction to publish all factor-by-variant cells. The plan was locked inside the repository before the designated confirmatory execution, but it was not registered with an external timestamping service and the research team already had access to the underlying data environment. It is therefore best described as a repository-locked confirmatory specification, not a strict prospective preregistration.

The run is bound to analysis commit `cbf5414c2613032dfa29ef2c295c760a3f4769ef`. The canonical receipt's integrity digest is

`8655e338433d01cbb518eab5fad68957baa7b422633e60294635c4a1c6df5f5c`, and the plan digest is `5682a806e1ec0702abddc49d500ba6518ffb3ee6d949700127e979055f7b1878`. The receipt also declares `performance_claim=false`, `generalization_claim=false`, and `usable_for_trading_decisions=false`. These fields are substantive constraints on the paper's language, not merely metadata.

Evidence chain used in the study



Each artifact is hash-bound; the public receipt records all planned cells and explicitly gates performance claims.

Figure 1. Hash-bound evidence chain and public claim boundary

5. Results

5.1 Complete pre-specified matrix

Table 3. All planned out-of-sample factor-by-variant results

Variant	Factor	Mean IC	NW t	Top-univ.	Mean N	Months
M0	ROE	0.0435	2.45	0.562%	4,533	18
M0	Mom. (60d)	-0.0514	-1.68	0.765%	4,516	18
M0	Low vol. (20d)	0.0529	1.86	-1.038%	4,527	18
M0	Composite	0.0302	1.38	0.502%	4,516	18
M1	ROE	0.0461	2.63	0.593%	4,554	18
M1	Mom. (60d)	-0.0495	-1.62	0.784%	4,538	18
M1	Low vol. (20d)	0.0550	1.94	-0.993%	4,549	18
M1	Composite	0.0332	1.53	0.539%	4,538	18
M2	ROE	0.0105	0.61	-0.111%	4,554	18
M2	Mom. (60d)	-0.0495	-1.62	0.784%	4,538	18
M2	Low vol. (20d)	0.0550	1.94	-0.993%	4,549	18
M2	Composite	0.0038	0.18	-0.007%	4,538	18
M3	ROE	-0.0108	-0.47	-0.393%	4,339	18
M3	Mom. (60d)	-0.0446	-1.86	0.864%	4,328	18
M3	Low vol. (20d)	0.0291	0.91	-1.417%	4,339	18
M3	Composite	-0.0194	-0.76	-0.340%	4,328	18

Note. The top-universe column is the equal-weight top signal quintile return minus the equal-weight universe mean over the 20-session horizon. It is not transaction-cost adjusted or self-financing. NW t uses lag three. No multiple-testing adjustment is applied.

5.2 Survivorship control: M0 to M1

Replacing the final-survivor universe with historical listing membership produces small rather than sweeping changes. ROE mean IC rises from 0.0435 to 0.0461, momentum becomes slightly less negative (-0.0514 to -0.0495), low-volatility IC rises from 0.0529 to 0.0550, and composite IC rises from 0.0302 to 0.0332. Mean cross-sectional sizes also increase by roughly 21–22 securities because historically listed names that do not survive to the terminal master re-enter the relevant dates.

This pattern does not imply that survivorship bias is unimportant in general. It says that, for these dates, fields, and summary statistics, the membership correction does not explain away the baseline patterns. Indeed, the modest increases show why the sign and size of a bias should be measured rather than assumed. M0 and M1 still share the more consequential accounting-timing convention, so neither provides a credible final estimate for ROE.

5.3 Publication timing: M1 to M2

Publication-date alignment is the decisive transition. ROE mean IC falls by 0.0356, from 0.0461 to 0.0105, equivalent to 77.2% of the M1 value. Its top-quintile-minus-universe spread moves from +0.593% to -0.111% per 20-session measurement window. Composite mean IC falls by 0.0294, from 0.0332 to 0.0038, an 88.5% decline, while its spread moves from +0.539% to -0.007% . These comparisons hold universe membership and the price-return convention fixed. They therefore isolate the effect of delaying the accounting input until publicationDate within the available data model.

Momentum and low volatility are identical in M1 and M2, down to the reported precision. That invariance is expected because both signals use only the price panel; their inputs are not governed by accounting publication dates. It is an internal implementation check: the timing change affects only the signal families that depend on the changed field. The result also clarifies the mechanism behind the composite deterioration. With 50% weight on ROE rank, the composite inherits a large portion of the accounting-timing sensitivity.

The M1 accounting convention effectively allows a fiscal-period ROE value to become known at the report-period end even when the company published it later. The M2 result demonstrates that this seemingly small timestamp choice is economically and statistically material. It should not be described as data-revision evidence, however. The experiment changes first availability, not the later values of distinct historical vintages.

5.4 Implementation stress: M2 to M3

M3 adds ST and suspension exclusions, a CNY 5 million 20-session mean amount floor, and a one-session lag. Average cross-sectional size falls from 4,554 to 4,339 for ROE (4.7%) and from 4,538 to 4,328 for momentum and the composite (4.6%). ROE mean IC changes from +0.0105 to -0.0108; composite changes from +0.0038 to -0.0194. Their top-quintile contrasts also become more negative, at -0.393% and -0.340%, respectively.

Price-signal results change less uniformly. Momentum mean IC becomes slightly less negative, from -0.0495 to -0.0446, while its positive top-quintile spread increases from 0.784% to 0.864%. Low-volatility mean IC falls from 0.0550 to 0.0291, while its already negative top-quintile spread becomes more negative, from -0.993% to -1.417%. Because M3 changes eligibility and return timing together, these shifts cannot be assigned to liquidity, special-treatment status, suspension, or lag separately. A later ablation study would need one variant per component.

Table 4. Key accounting-signal transitions

Factor	Transition	Mean IC	Top-universe	Interpretation
ROE	M1→M2	0.0461 → 0.0105	+0.593% → -0.111%	Publication timing; clean within design
Composite	M1→M2	0.0332 → 0.0038	+0.539% → -0.007%	Publication timing inherited through 50% ROE weight
ROE	M2→M3	0.0105 → -0.0108	-0.111% → -0.393%	Filters plus lag; combined effect
Composite	M2→M3	0.0038 → -0.0194	-0.007% → -0.340%	Filters plus lag; combined effect

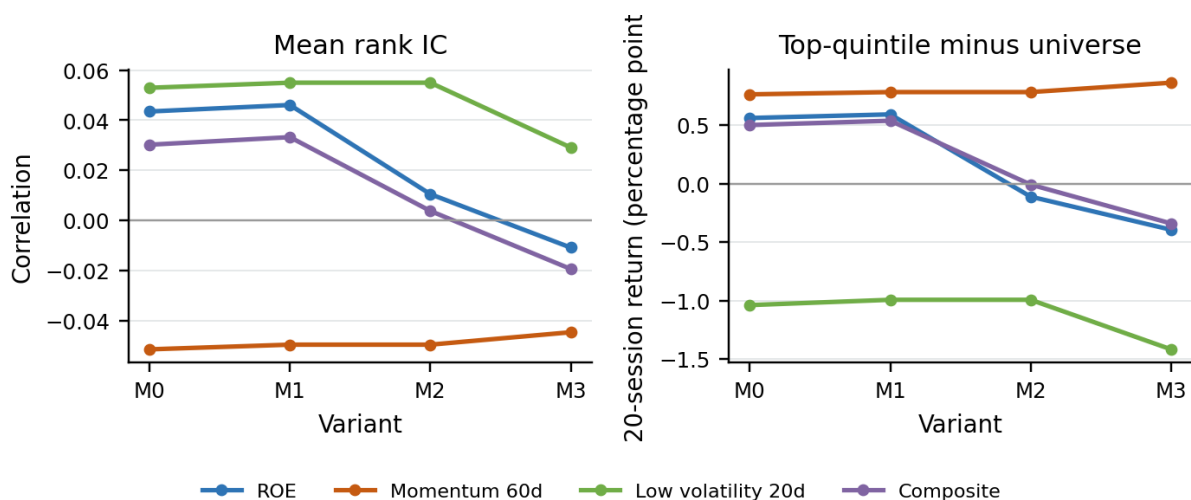


Figure 2. Metric paths across cumulative audit variants

5.5 Why the two metrics disagree

Across all four variants, momentum has a negative mean rank IC but a positive top-quintile-minus-universe return. Low volatility has a positive mean rank IC but a negative top-quintile contrast. The two facts are not logically inconsistent. Rank IC uses the ordering of the entire eligible cross-section and is sensitive to broad monotonic association. The quintile statistic examines only the top signal bin relative to the whole universe and can be dominated by curvature, tail behavior, unequal dispersion across dates, or a concentration of extreme outcomes. Averaging the monthly statistics separately can further detach their signs.

For this reason, the paper does not relabel a factor according to its favorable metric. A positive top-quintile contrast cannot rescue a negative full-rank association without a pre-specified portfolio construction and additional diagnosis. Conversely, a positive IC does not imply that a top-bin strategy earns a positive premium. The disagreement is itself a result: it identifies where monotonic factor narratives fail and where future work should inspect decile profiles, winsorization, sector composition, and month-by-month contributions.

5.6 Statistical interpretation

Several Newey–West t-statistics in M0 and M1 exceed two in absolute value, including ROE at 2.45 and 2.63. Those values weaken to 0.61 in M2 and –0.47 in M3. The apparent strength of the naïve ROE signal therefore depends on the timestamp convention. Yet even the larger values should not be treated as confirmatory discoveries. There are only 18 monthly observations, four correlated signals, four cumulative variants, two headline metrics, and no multiple-testing correction. The variant matrix is designed for transparency, not for maximizing a test statistic.

The appropriate inference is comparative. Relative to the same code path and universe, publication timing removes most of the average ROE and composite IC. That within-study attenuation is more credible than a claim that any individual post-audit mean is nonzero. A longer prospective holdout, a small-sample robust inference design, and independent data reconstruction would be required for stronger generalization.

6. Research Governance and Reproducibility

6.1 Canonical evidence instead of report accumulation

Before the current evidence chain, the repository contained multiple local reports with inconsistent state descriptions. Some narrative artifacts described the project as portfolio-ready, while the latest readiness checks reported missing registry entries and candidate results. Such inconsistency is

predictable when prose files are treated as independent authorities. The repair is architectural: one machine-readable public status points to one canonical receipt, and legacy documents are explicitly labeled as non-authoritative.

The receipt is generated from the run and contains the planned matrix rather than a manually copied highlight. The public status file exposes the evidence level and the claim gates. This design makes contradictions testable: automated checks can fail if documentation asserts trading readiness while the canonical fields remain false, if a required cell is missing, or if a referenced plan or code hash no longer matches. Research governance thus becomes part of the software contract.

6.2 Reproducibility levels

Reproducibility is not binary. At the software level, the repository exposes the analysis logic, schema contracts, fixtures, and tests. At the identity level, the receipt records hashes and row counts for the non-redistributable export. At the empirical level, an authorized researcher with the same data can rerun the study and compare the result matrix. At the fully independent level, a reader would need an independently sourced and historically versioned dataset; that level is not achieved here.

This layered statement follows the spirit of reproducible computational research articulated by Sandve, Nekrutenko, Taylor, and Hovig (2013): record inputs, preserve the analysis trail, and make computational steps inspectable. The paper adds a finance-specific requirement: the record must include the historical information set and execution clock, because a perfectly reproducible look-ahead backtest remains wrong.

6.3 Data mining controls

The repository contains many strategy variants outside this study. Selecting the highest historical Sharpe ratio or best-looking chart from that menu would invite a data-mining critique. The confirmatory study therefore defines a small factor set with fixed composite weights and reports all 16 cells. No factor is removed when its sign is inconvenient, no variant is renamed as the primary model after the run, and no result is promoted as a trading candidate.

This rule does not erase earlier exploration. The team had access to the research environment before the plan was locked, and the signals are standard rather than newly conceived. The value of locking is narrower: it prevents the final published table from becoming another best-of-many selection exercise. Stronger future work should commit the next specification to an external timestamp before acquiring the prospective holdout.

7. Limitations and Next Research Steps

First, the out-of-sample time series is short. Eighteen monthly observations cover multiple market conditions but are insufficient for precise long-horizon inference, tail analysis, or stable HAC estimation. The wide cross-section improves the measurement of each month's association, yet it does not replace independent time-series observations. A two- to three-year prospective extension would be more informative than adding more variants to the existing window.

Second, the raw export is licensed and unavailable in the public repository. Hashes support identity checks but not unaided replication. The public fixtures demonstrate schemas and timing logic, not the reported market results. A useful next step is to define an adapter for an independently licensed or openly redistributable data source and verify whether the same attenuation appears.

Third, `publicationDate` is not a complete vintage system. The study cannot observe all corrections, restatements, vendor backfills, or the exact intraday release time. A date-level release may still be optimistic if information was published after the close but used at that close. M3's one-session lag reduces that risk, but the underlying revision history remains unknown. The paper therefore avoids the claim that it has studied data revisions.

Fourth, M3 is a bundled stress test. It jointly changes eligibility filters and entry timing, so its deterioration cannot be assigned to a single mechanism. A pre-specified ablation should create separate variants for ST exclusion, suspension handling, liquidity threshold, next-open or VWAP entry, and return-horizon alignment. Those variants should be treated as a new study rather than appended opportunistically to the current matrix.

Fifth, the outcome is not a portfolio. The top-quintile contrast uses equal-weight security returns and subtracts the universe mean; it does not model a financed short leg, turnover, commissions, stamp duty, slippage, price limits, borrow availability, sector exposures, market beta, position limits, or capacity. The receipt correctly sets `usable_for_trading_decisions` to false. Portfolio readiness would require a separate protocol and a new evidence class.

Sixth, factor definitions are intentionally simple. ROE is vendor supplied; momentum and volatility use fixed lookbacks; the composite weights are heuristic. No industry neutralization, size residualization, outlier treatment, or accounting quality screen is applied. These choices enhance transparency but may confound economic mechanisms. Any refinement should be motivated prospectively and evaluated on fresh data.

Seventh, the study does not offer a causal explanation for why publication alignment weakens ROE. The attenuation is consistent with look-ahead contamination under report-period timing, but it may also

reflect the age and distribution of filings, sector composition, missingness, or vendor field construction. Filing-lag distributions and month-level attribution would help distinguish these channels.

Finally, the locked specification is repository-local rather than externally preregistered. It improves completeness and auditability but cannot certify that no prior result influenced the choice of signals. The strongest next design is prospective: publish a signed plan, fix the raw-data interface, wait for new monthly observations, and release the receipt automatically without discretionary model selection.

8. Conclusion

This study replaces an ambiguous collection of backtest artifacts with a narrower and more defensible empirical claim. In a large A-share cross-section, reconstructing historical listing membership changes familiar factor statistics only modestly, while aligning ROE to publication dates removes most of the naive ROE and composite rank association. Adding tradability filters and a one-session lag pushes both accounting-dependent signals below zero in this short holdout. Metric discordance for momentum and low volatility further warns against selecting whichever summary looks most favorable.

The result is not a failed strategy search; it is a successful evidence audit. The repository now distinguishes real-market out-of-sample statistics from performance claims, binds the empirical output to identifiable data and code, and publishes the entire planned matrix. Those controls do not solve proprietary data access, vintage reconstruction, transaction costs, or external validity. They do make the remaining uncertainty visible.

For open-source quantitative research, visibility is the standard worth optimizing. A working paper should state what was measured, what information was allowed, which alternatives were run, what is missing, and which claims remain prohibited. On those terms, the project is research-ready for scrutiny, not portfolio-ready for deployment.

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